

Chapter 4: Atomic Structure & Bonding

Quantum Numbers, Electronic Configurations, Periodic Trends & Chemical Bonds

4.1 Atomic Structure & Quantum Numbers

An atom consists of a dense nucleus (protons and neutrons) surrounded by electrons in quantum orbitals defined by four quantum numbers:

1. Principal quantum number (n): Energy level.
2. Angular momentum quantum number (l): Subshell shape (s, p, d, f).
3. Magnetic quantum number (m_l): Orbital orientation.
4. Spin quantum number (m_s): Electron spin ($+\frac{1}{2}$, $-\frac{1}{2}$).

4.2 Periodic Trends

Properties of elements vary systematically across the periodic table:

- **Atomic Radius:** Decreases across a period, increases down a group.
- **Ionization Energy & Electronegativity:** Increase across a period, decrease down a group.
- **Electron Affinity:** Becomes more exothermic across a period.

4.3 Chemical Bonding: Ionic, Covalent & Metallic

Atoms bond to achieve stable valence shell configurations (octet rule):

- **Ionic Bonding:** Complete transfer of electrons between metals and non-metals.
- **Covalent Bonding:** Sharing of electron pairs between non-metals, categorized by polarity and hybridization.
- **Metallic Bonding:** A 'sea of electrons' surrounding positive metal ions, conferring high electrical and thermal conductivity.